# Operations Runbook (SOPs)

**Ref:** 5.6
**Product:** iConnect Plus
**Organisation:** Cache Digitech
**Classification:** Internal — Confidential
**Version:** 2.0
**Date:** 2026-11-03
**Deployment model:** On-premises (Cache Digitech premises) · VPN for remote access
**Audience:** Platform on-call, IT operations

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## 1. Purpose

Standard operating procedures for **deploy**, **restart**, **patch**, and **on-call** operations for iConnect Plus on the **internal Docker Compose** host at Cache Digitech premises.

## 2. Prerequisites

| Item | Requirement |
|---|---|
| Access | SSH to host via LAN or **VPN** |
| Repo path | **Template / Pending sign-off** (e.g. `/opt/Cache-ERP`) |
| Compose files | `docker-compose.yml`, `docker-compose.infra.yml`, `docker-compose.app.yml` |
| Secrets | Production `.env` on host only |
| Example URL | `https://172.16.200.30:8443` (internal) |

## 3. Service map

| Service | Container | Depends on |
|---|---|---|
| Edge | `erp-nginx` | web, api |
| Admin UI | `erp-web` | api |
| ESS | `erp-employee-app` | api |
| API | `erp-api` | postgres, redis, rabbitmq, minio |
| Worker | `erp-celery-worker` | postgres, redis, rabbitmq |
| Beat | Celery beat (if deployed) | rabbitmq |
| DB | `erp-postgres` | — |
| Cache | `erp-redis` | — |
| Broker | `erp-rabbitmq` | — |
| Objects | `erp-minio` | — |
| AV | `erp-clamav` | — |

## 4. SOP — Health check

```
cd /opt/Cache-ERP   # adjust path
docker compose ps
curl -k -s -o /dev/null -w "%{http_code}" https://127.0.0.1:8443/
docker logs --tail 50 erp-api
docker logs --tail 20 erp-celery-worker
```

| Check | Healthy signal |
|---|---|
| All containers | Up (or `healthy` if defined) |
| HTTPS | HTTP 200/302 |
| API | No repeated 5xx in logs |

Platform metrics route (if enabled): internal SLO endpoint — **Template / Pending sign-off**.

## 5. SOP — Rolling restart (no schema change)

| Step | Command / action |
|---|---|
| 1 | Notify users if broad impact |
| 2 | `docker compose restart erp-celery-worker` |
| 3 | `docker compose restart erp-api` |
| 4 | `docker compose restart erp-web erp-employee-app` |
| 5 | `docker compose restart erp-nginx` |
| 6 | Run health check (§4) |

Order preserves API before UI; adjust if maintenance window requires full stop.

## 6. SOP — Full stack stop / start

```
docker compose down          # caution: does not remove volumes
docker compose up -d
```

**Never** `down -v` in production without DR approval (destroys volumes).

## 7. SOP — Deploy new release

| Step | Action |
|---|---|
| 1 | Tag / commit recorded in 2.4 change record |
| 2 | Pull Git at release tag on host |
| 3 | Build images: `docker compose build` (or load from internal registry) |
| 4 | Run Alembic: `docker compose run --rm erp-api alembic upgrade head` |
| 5 | Restart api → worker → web → nginx |

| | |
|---|---|
| 6 | Smoke test: login, one critical workflow |
| 7 | Monitor logs 30 min |

Rollback: redeploy previous image tag; Alembic downgrade only if approved and tested.

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## 8. SOP — Database migration only

| Step | Action |
|---|---|
| 1 | Backup DB (4.4) |
| 2 | Run `alembic upgrade head` |
| 3 | Verify migration version table |
| 4 | Application smoke test |

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## 9. SOP — OS / Docker patch

| Step | Owner |
|---|---|
| Schedule maintenance window | IT + Platform |
| Apply host patches | IT |
| Reboot if required | IT |
| Start Compose stack | Platform |
| Verify 5.2 time sync post-reboot | IT |

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## 10. SOP — Secret rotation

See **4.2**. Typical order: generate new secret → update `.env` → restart affected services → invalidate sessions if JWT key changed.

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## 11. SOP — Disk space emergency

| Step | Action |
|---|---|
| 1 | `df -h` on host |
| 2 | Trim old logs per 5.1 retention |
| 3 | Purge unused Docker images `docker system prune` (careful) |
| 4 | Expand volume or archive backups |
| 5 | If PostgreSQL full — stop writes; extend disk |

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## 12. On-call

| Item | Detail |
|---|---|
| Primary | **Template / Pending sign-off** |
| Secondary | **Template / Pending sign-off** |
| Escalation | IT → Platform lead → IC (5.4) |
| Ticket queue | **Template / Pending sign-off** |

### 12.1 On-call first 15 minutes

1. Acknowledge alert.
2. Check container status and disk.
3. Review recent deploys (2.4).
4. Mitigate (restart, scale, failover per 5.5).
5. Open incident record if S1/S2.

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## 13. Related documents

| Ref | Document |
|---|---|
| 4.4 | Backup & Restore |
| 5.5 | BCP / DR Plan |
| 5.3 | Hardening Baseline |
| 2.4 | Change & Release Records |

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## 14. Change log

| Date | Change | By |
|---|---|---|
| 2026-11-03 | v2.0 Compose runbook | Platform |